

Organic

- Carbon dioxide, carbon monoxide, and all carbonates in general are not considered organic compounds.

* Carbon is a non-metal in group (IV). It forms covalent compounds; its valence is 4 so it forms 4 bonds.

→ Carbon exists in several forms:

- 1- diamond
- 2- Graphite
- 3- fullerene

3 special features of the covalent bonding involving carbon:-

- 1- Carbon atoms join to each other to form long chains: straight and branched, atoms of other elements can then attach to the carbon chain.
- 2- Carbon atoms can be linked by single, double, or triple bonds.
- 3- Carbon atoms can also attach themselves in rings.

* Homologous series: A family of similar compounds with similar chemical properties due to the presence of the same functional group.

* Functional group: An atom or group of atoms in a molecule that determines the chemical properties of a homologous series.

General characteristics of a homologous series:

- 1- They have the same general formula.
- 2- They have the same functional group.
- 3- They share same chemical properties because they have the same functional group.
- 4- Each member differs by a $-CH_2-$ unit as the chain gets longer.
- 5- Show a trend in their physical properties such as:
 - Melting & boiling point increase
 - Viscosity increase → Flow less easily
 - Flammability decrease → Burn less easily

as the chain gets longer

Hydrocarbons (HC): Compounds that contain carbon and hydrogen atoms only.

Saturated HC:

Contain single C-C bonds only. Known as Alkanes

Unsaturated HC:

Contain double or triple C=C / C≡C bonds.

Alkenes

C=C

Alkyne

C≡C

Alkanes: are saturated hydrocarbons that contain only single C-C bonds.

* They occur naturally in petroleum and natural gas.

↳ Contains alkanes with up to 70 carbons

↳ Mainly methane, with small amounts of -ethane -butane -Propane

* General formula: C_nH_{2n+2}

- 1 carbon: meth-
- 2 carbons: eth-
- 3 carbons: Prop-
- 4 carbons: But-

- * Empirical Formula: The simplest possible whole number ratio of atoms in a compound.
- * Molecular Formula: Represents the actual number of atoms in a molecule.
- * Structural Formula: Shows each unit in the structure. No bonds except for C=C
- * Displayed Formula: Shows all the atoms in the molecule and how they are bonded together

→ As the length of the hydrocarbon chain increases, the strength of the intermolecular forces (weak forces between the molecules) is increased; this explains the increase in melting & boiling points

Physical properties of alkanes:

- The first four alkanes are colorless gases
- $C_5 - C_{17}$ are colorless liquids
- $C_{18} \leq$ are colorless solids.
- * All alkanes are insoluble in water.

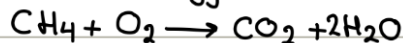
Chemical properties:

- 1- Because they have single C-C covalent bonds, they are called saturated.
- 2- They are unreactive compounds; they do not react with steam, acids or bases.
- 3- Short-chain alkanes burn very well in a good supply of oxygen, forming CO_2 and water vapour, and they give out plenty of heat. So, they are used as fuels.
- 4- If there is not enough oxygen, alkanes undergo incomplete combustion, giving poisonous carbon monoxide.
- 5- They cannot undergo addition reaction, but they undergo substitution reaction.

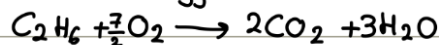
Reactions of alkanes:

- 1- Combustion: * Alkanes burn very exothermically, so they are used as fuel
 - * When they burn in sufficient supply of oxygen, they produce carbon dioxide and water.
 - * When they burn in insufficient supply of oxygen, they produce carbon monoxide, and sometimes soot.

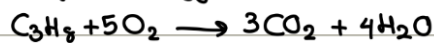
- Methane + Oxygen → Carbon dioxide + Water



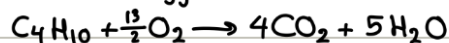
- Ethane + Oxygen → Carbon dioxide + Water



- Propane + Oxygen → Carbon dioxide + Water



- Butane + Oxygen → Carbon dioxide + Water



2- Substitution reaction with halogens: Alkanes react with chlorine in sunlight; chlorine displaces hydrogen atom.

→ One atom or group of atoms is replaced by another atom or group of atoms

* Condition: Ultraviolet light / sunlight → To provide the activation energy (E_a) for the reaction.

If we use excess chlorine gas, all the hydrogen atoms will be substituted.

* $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ → Chloromethane

* $CH_3Cl + Cl_2 \rightarrow CH_2Cl_2 + HCl$ → Dichloromethane

* $CH_2Cl_2 + Cl_2 \rightarrow CHCl_3 + HCl$ → Trichloromethane

* $CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$ → Tetrachloromethane

Can be used as solvents, but they cause health problems, so they are used less and less.

Acidic gas that turns damp litmus paper red.

Uses of alkanes: - As fuel

- Manufacturing of alkenes (catalytic cracking)
- Waxes and lubricating oil.

Structural isomers: Compounds with the same molecular formula, but different structural formula.

→ Branched

→ Unbranched (straight)

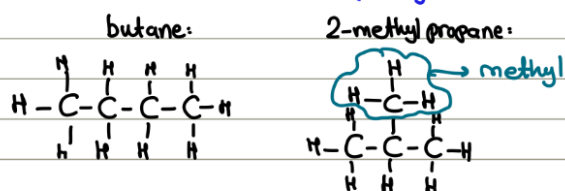
→ Methane, ethane, propane do not have isomers

→ Butane has two isomers.

Since they have different structures, they have different chemical properties:

→ Branched isomers have lower boiling points than straight chain isomers because molecules with branched chains cannot get so close together. So the attraction between them is less strong and less heat is needed to overcome it.

→ Branched-chain hydrocarbons burn less easily than their straight-chain isomers. So they are used in petrol to control combustion and stop engine.



Haloalkanes: - F = Fluoro-

- Cl = Chloro-

- Br = Bromo-

- I = iodo-

Obtaining alkanes: (Fossil fuels):

Three types of fossil fuels:

1- Natural gas (colorless gas): Mainly methane CH_4

2- Petroleum (crude oil): A mixture of many types of alkanes having different lengths of carbon chains

3- Coal (Black solid): Mainly carbons

Fossil fuels contain some sulfur impurities.

Fossil fuels are non-renewable resources

* We can't use petroleum as fuel is a sticky black liquid that is difficult to set alight; when it burns, it produces a cloud of poisonous black smoke

* In oil refining, the petroleum is separated into smaller groups; each group is with a limited range of carbon atoms called fractions.

* The hydrocarbon fractions are separated by fractional distillation, according to different boiling points of each fraction.

→ Each petroleum fraction is a group of molecules of a similar size that have similar boiling points.

* As the chain length increases, the boiling points of the compound rises.

* If you heat petroleum enough, compounds will boil to form gases. If these rise up a tall tower, they will cool as they rise and condense at different heights in the tower depending on their boiling points.

Small molecules:

- Low boiling points
- Higher volatility
- Light in color
- More Flammable
- Lower viscosity (Flow easily)

Large molecules:

- High boiling points
- Lower volatility
- Dark in color
- Less Flammable
- Thick / viscous.

The trends the fractions show as you go down.

- The higher its boiling points
- Less volatile
- More viscous
- Less Flammable

Fraction name:

- Refinery gas
- Gasoline (petrol)
- Naphtha
- Kerosene (paraffin)
- Diesel oil
- Fuel oil (heavy gas oil)
- Lubricating oil
- Bitumen

Uses:

- Bottled gases for heating and cooking
- Fuels for cars
- Starting point or feed stock for many chemicals and plastics
- Fuel for jet aircrafts, oil stoves, and lamps
- Fuel for lorries and tractors. (diesel engines)
- Fuel for power stations, ships, and home central heating
- Lubricants, waxes and polishes.
- Making road surfaces and sealing oils

* All fractions are used as fuels except for naphtha, lubricating oil and bitumen.

* The liquid fuel fractions may contain sulfur. This must be removed because sulfur burns in air to give sulfur dioxide gas, which causes acid rain.

→ Demand for the various fractions from the refinery doesn't match their supply for oil.

→ The demand for the lighter fractions (petrol / kerosene) is greater than the demand for the heavier fractions. (paraffin / diesel)

* Larger molecules can be broken down into smaller more valuable fractions by cracking.

Alkenes: - Unsaturated hydrocarbons: molecules contain $C=C$ double covalent bonds.

- The functional group of alkenes is $C=C$

* They are unsaturated because it is possible to break the double bond and add extra atoms to the molecule

- The general formula is C_nH_{2n}

* The simplest alkene is ethene (C_2)

* There is no methene.

* Cycloalkanes are isomers of alkenes.

Properties of alkenes:

- The first four alkenes are colorless gases

- C_5 to C_7 are colorless liquids

- C_{18} are colorless solids

* All alkenes are insoluble in water

* Alkenes are more reactive than alkanes, because the double bonds can be broken to add on other atoms.

Reactions of alkenes:

1- Combustion: They burn in excess oxygen with a smoky yellow flame to produce carbon dioxide and water.



2- Addition reaction: - Two reactants add together to form one product only.

- The double bonds can be easily broken, allowing other atoms to bond.

- The addition reaction turns an unsaturated alkene to a saturated compound.

a) Addition of hydrogen: Hydrogenation: Hydrogen reacts with alkenes to form alkanes.

Conditions: Temperature: $60^\circ C$



Catalyst: Nickel

* This reaction is classed as a Redox reaction

→ Gaining hydrogen is reduction

* This reaction is used to make margarine from unsaturated vegetable oils.

→ Losing hydrogen is oxidation.

b) Reaction of alkene with bromine (addition of bromine water / aqueous bromine):



→ The color will change from orange-brown to colorless. This is because the alkenes decolorize bromine

Test for unsaturation:

1- With alkenes, the color changes from orange to colorless.

2- With alkanes, the color remains orange-brown because alkanes do not undergo addition reactions.

c) Reaction of alkene with steam: Hydration: Catalytic addition of steam:



Conditions: - Temperature = $300^\circ C$

- Pressure = 60 atm / 6000 kPa

- Catalyst = concentrated phosphoric acid (H_3PO_4)

* The reverse of hydration is dehydration: Heat with a dehydrating agent: concentrated H_2SO_4 , by passing ethanol vapour over hot aluminum oxide (catalyst).

Making alkenes:

Cracking: * Alkenes do not occur naturally in petroleum. They are made from alkanes by cracking.
↳ It's the breaking of larger chains of hydrocarbons into smaller, more useful hydrocarbons using heat and a catalyst.

(Thermal decomposition of alkanes): Conditions: - 500°C

- Catalyst: Aluminium oxide.

Products of cracking:

1- Smaller chain alkanes to supply demand for short alkanes (fuel)

2- Alkene: - To form polymers

- To make alcohol

3- Hydrogen gas: - To make ammonia

- In hydrogen-oxygen fuel cell

- In rockets as fuel.

* The high temperature and catalyst are needed because you have to break very strong carbon-carbon covalent bonds in the alkane molecule

Reasons of cracking:

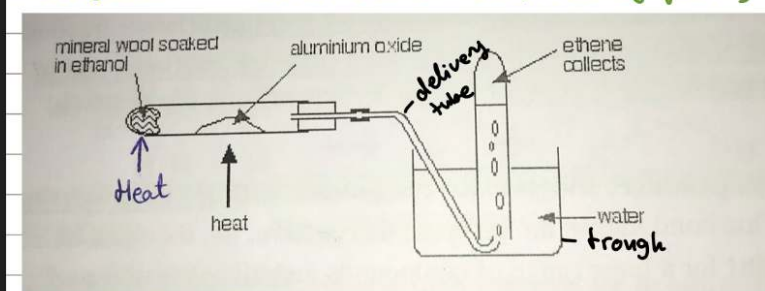
1- Helps you make the best use of petroleum

2- Cracking always produces short chain compounds with carbon-carbon double bonds. This bond makes the compounds reactive. So they can be used as a starting point for a huge range of compounds including plastics and medical drugs.

3- The smaller alkanes are used for motor fuels which burn more efficiently

Preparation of ethene in Labs:

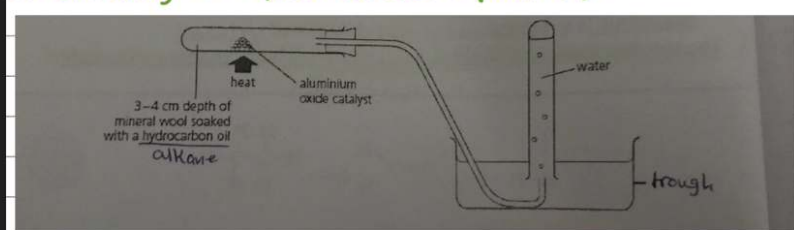
1- Dehydration of ethanol: Can be accomplished by passing alcohol vapour over a heated aluminium oxide catalyst.



Why do we use ceramic wool?

To hold the alcohol at the end of the tube

2- Cracking of liquid alkane (paraffin):



* Paraffin is absorbed into a ceramic wool to hold it at the end of tube.

* As paraffin vapour passes over the hot catalyst: aluminium oxide, it is converted (cracked) into a smaller chain alkane and ethene gas. Ethene is collected over water (downward displacement of water since ethene is insoluble in water).

- The delivery tube must be removed before you stop heating to prevent the suck back of water which causes the test tube to break.

- The first portion collected must be thrown because it is simply air

Alcohol is a homologous series of compounds that contain a hydroxyl group (-OH) as their functional group.

General formula: $C_nH_{2n+1}OH$ / $C_nH_{2n+2}O$

Alkane + ol $\rightarrow$ Alcohol

The simplest alcohol contains one carbon: methanol

The early alcohols are neutral, colorless liquids that do not conduct electricity.

Preparation of ethanol: C_2H_5OH

1- Hydration of ethene / adding reaction of steam:



Conditions: 1- temperature = $300^\circ C$

2- Pressure = 60 atm / 6000 Kpa

3- Catalyst: concentrated phosphoric acid.

The reaction is reversible and exothermic. High pressure and low temperatures would give the best yield. But a moderate temperature is used for decent rate.

- A catalyst is used to speed up the reaction, so it is called catalytic addition.

- Ethanol produced is pure.

2- Fermentation of sugar: Ethanol is made from the glucose using yeast in the absence of air. The enzymes in the yeast act as catalysts.

- The reaction is exothermic



Conditions: Pressure = 1 atm

- Temperature = $25^\circ - 35^\circ C$

- No air is allowed (anaerobic respiration)

- Yeast to produce enzymes.

* The air lock allows CO_2 to escape but prevents airborne bacteria from entering; otherwise ethanol is oxidized into ethanoic acid.

Observations:

1- Bubbles (of CO_2)

2- Solution turns cloudy / turbid (due to the multiplying yeast)

3- Rise in temperatures (because it is an exothermic reaction).

* Ethanol is toxic to yeast; so fermentation is a self-limiting. Once the concentration of ethanol reaches above 15% / or the sugar runs out, the yeast will die and the fermentation stops.

* Ethanol produced is impure, needs to be purified by fractional distillation.

	Hydration of ethene catalytic addition of steam	Fermentation
Equipment	dis complex set up required expensive	adv simple equipment needed
Raw materials	dis uses non-renewable resources (crude oil)	adv uses renewable resources (sugar cane)
Type of process	adv continuous process - a stream of reactants is constantly passed over a catalyst	dis batch process - everything is mixed together in a reaction vessel and left for several days and the process repeated again
Rate of reaction	adv fast	dis very slow (several days)
Quality of product	adv produces pure ethanol	dis produces a dilute solution requiring further processing
Atmospheric effects	adv no greenhouse gases produced but pollutants are formed from the burning of fossil fuels to maintain high temperatures	dis carbon dioxide produced which is a greenhouse gas
Reaction conditions	dis high temperatures and pressures required increasing the energy input and cost	adv low temperatures required

Properties of ethanol:

1- Volatile; evaporates easily. Boiling point = $78^\circ C$. Used in: - Glues, - perfumes
- Printing inks. - Aftershave

2- Flammable liquid

3- Burns in air with a blue flame to form CO_2 and H_2O and plenty of heat $\rightarrow$ Used as fuel

4- Neutral to litmus paper.

Reactions of ethanol:

1- Burning: (ethanol is used as a fuel):



2- Oxidation

Alcohol $\xrightarrow{[\text{O}]}$ Carboxylic acid + Water

a) Acid Fermentation / bacterial oxidation. **SLOW**

b) Heating with a powerful oxidizing agent: Acidified potassium manganate(VII) (KMnO_4)
↳ Sulfuric acid is added.

Ethanol oxidation Ethanoic acid + Water



3) Dehydration of ethanol to produce ethene: Ethanol can be dehydrated by passing the alcohol vapour over heated aluminium oxide catalyst.

4) Esterification: Reacts with carboxylic acid to form sweet-smelling oily liquids known as esters.

Uses of ethanol:

- As a solvent
- As a fuel (biofuel)

To test for alcohols:

* Heat the alcohol is acidified potassium manganate(VII). The color changes from purple to colorless. This is because alcohol is a reducing agent that will reduce potassium manganate.

Carboxylic acids form a homologous series of organic compounds with a carboxyl ($-\text{COOH}$) group as their functional group.

General formula: $\text{C}_n\text{H}_{2n+1}\text{COOH}$

Alkane + -oic $\rightarrow$ alkanic acid

Formation of ethanoic acid from ethanol:

Ethanol can be oxidized to ethanoic acid in two ways:

1- Acid fermentation / Bacterial oxidation: Oxidation in air in the presence of a bacteria

* This reaction takes place in the presence of oxygen.



2- Using oxidizing agents: Ethanol is oxidized much faster by warming it with a powerful oxidizing agent: potassium manganate (VII), in the presence of an acid. (to make it acidified). The color changes from purple to colorless.

Properties of carboxylic acids:

1- They can be made by the oxidation of alcohols

2- Their solutions turn damp litmus paper red.

3- They are weak acids; they're partially ionized in aqueous solutions. $[\text{H}^+]$ is low.

Reactions of ethanoic acid:

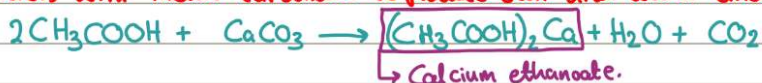
1- Reacts with any metal above hydrogen in the reactivity series to produce salt and hydrogen.



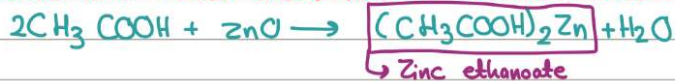
2- Reacts with ^{base}alkali to produce salt and water



3- Reacts with metal carbonate to produce salt and water and carbon dioxide.



4- Reacts with metal oxides to form salt and water. but the reaction requires heating because the E_a is high



* They form salts called alkanoate

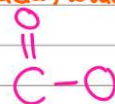
* The reaction of ethanoic acid is slower than any strong acids such as HCl , H_2SO_4 , HNO_3 ; because it is weak, partially ionized so less concentration of Hydrogen ions (H^+), and so less frequent collisions.

Differentiations between ethanol and ethanoic acid

Test	Ethanol	Ethanoic acid
Litmus paper	No color change (neutral)	Turns red
Sodium carbonate	No reaction	Effervescence of CO_2
Potassium manganate(VII)	Purple $\rightarrow$ colorless	No color change (remains purple), no further oxidation
Lightened splint	Catches fire	Doesn't catch fire

Esters: - A homologous series of organic compounds that contain an ester link ($-\text{COO}-$) as a functional group.

- Formed by heating an alcohol and carboxylic acid with concentrated sulfuric acid; which is called esterification condensation.



Condensation reaction: The reaction that occurs when two smaller organic molecules combine to form a more complex molecule, accompanied by the loss of small molecules such as water.

The reverse of condensation is hydrolysis.

Describe how to make ethyl ethanoate. Write an equation. (4 marks)

Heat ethanol with ethanoic acid with concentrated sulfuric acid.



To name esters: - Name the alcohol part: alkyl

- Then name the acid part: alkanate

Alkyl alkanate.

* What comes before the $-\text{COO}$ bond is the acid, after is the alcohol.

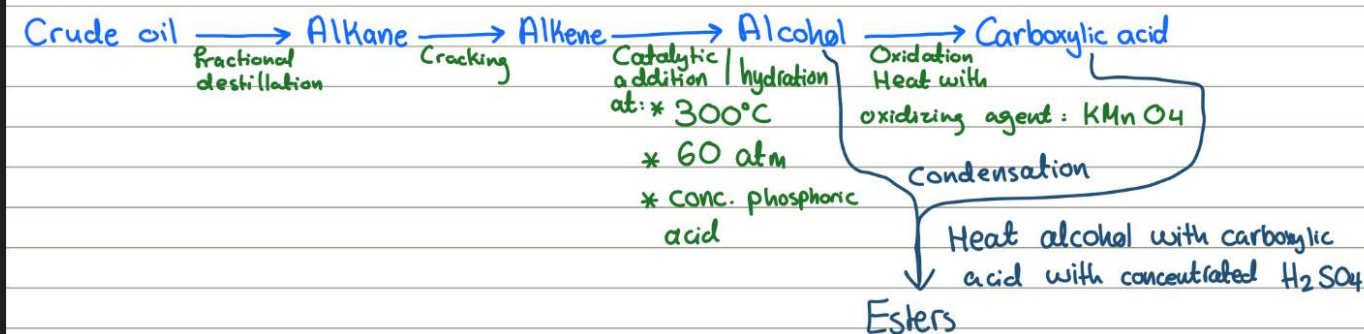
General Formula: $\text{C}_n\text{H}_{2n}\text{O}_2$

Properties of esters:

- 1- The simplest esters are colorless liquids which are immiscible with water
- 2- They have pleasant fruity smell
- 3- Artificial Flavouring are mixtures of esters which are added to soaps, shampoos, and perfumes
- 4- All esters contain ester link
- 5- Fats and vegetable oils are natural esters

Uses:

- Food Flavouring
- Perfumes, Shampoos, Soaps and aftershave



The more number of atoms, the more energy is released

Polymers: Large molecules, made up of small repeating units called monomers joined together by a process called polymerisation.

2 types of polymerization:

1- Addition polymerization: When catalysed and heated under pressure, unsaturated alkenes link together when the double bonds open. The spare carbons on each carbon atoms of the double bond is used to join up the molecules.

- One product only is formed.



* The formation of links between smaller monomer molecules containing $C=C$ double bonds to form a long chain macro molecule.

* All alkenes possess addition polymerisation to make plastics.

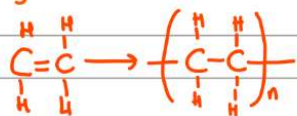
Conditions for polymerization:

1- High temperature

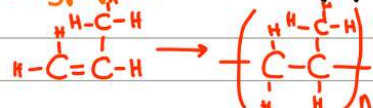
2- High pressure

3- Catalyst

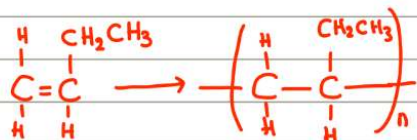
1- Polyethene: Monomer: Ethene



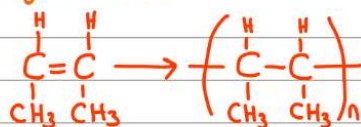
2- Polypropene: Monomer: propene



3- Poly but-1-ene: Monomer: but-1-ene



4- Poly but-2-ene: Monomer: but-2-ene



2- Condensation Polymerization: No double bonds break: Two different monomers join, each with a different functional group.

The monomers join at their functional group by getting rid of the small molecules.

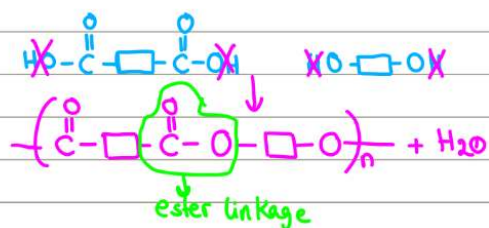
- Two products form: * Polymer

* Water

1- Synthetic macromolecules:

* **Polysters:** PET/terylene: They are polymers with ester linkage $(-COO)(-\overset{O}{\parallel}C-O)$

Monomers: - Diacid - diol



* Polysters are widely used in making water bottles, clothing, bed linen

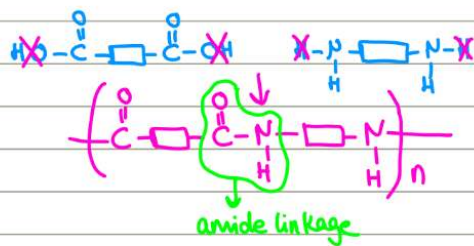
* Polysters can be broken down into monomers again by adding steam at high pressure and temperature. $\rightarrow$ Hydrolysis

Category	Details
Advantages of Plastic	- Cheap and easy to make - Lightweight (lighter than metals, wood, or glass) - Unreactive (does not corrode in air or water and is not affected by acids or alkalis) - Good insulator (does not conduct heat or electricity) - Durable (does not break easily when dropped) - Resistant to fire (does not catch fire easily but melts when heated) - Can be recycled (can be melted and molded into any shape)
Disadvantages of Plastic in Landfill Sites	- Non-biodegradable (cannot be broken down by microorganisms in soil or water) - Causes visual pollution (builds up and creates a mess) - Blocks drainage systems (clogs up drains and sewers, leading to flooding) - Releases toxic chemicals (when breaking down, some plastics release harmful substances that can contaminate soil and rivers)
Disadvantages of Plastic in the Ocean	- Harms marine life (fish and other animals mistake plastic for food, leading to starvation) - Photo degradation (breaks into microparticles when exposed to sunlight) - Toxic accumulation (marine animals ingest microbeads and microparticles, which can damage their health and fertility) - Creates large plastic waste patches (ocean currents carry plastic waste, forming huge floating islands of plastic)

Ways to Tackle Plastic Pollution	Details	Disadvantages of Recycling Plastics	Details
Make plastic easier to degrade (biodegradable)	- Develop plastics that break down more easily, e.g., cornstarch-based plastics. - Manufacture plastics that degrade with sunlight exposure. - Concern: If plastics break down into microplastics, it remains a problem.	Difficult separation	- Plastics have different physical properties, making recycling complex.
Reuse	- Refill plastic water bottles instead of buying new ones.	Sorting issues	- Plastics must be categorized into useful types, which is challenging.
Reduce	- Minimize plastic use where possible.	Manual labor required	- Lack of automation means plastics may need to be sorted by hand, increasing costs.
Burn with care	- Plastics can be burned in incinerators where toxic gases are trapped. - Heat produced can be used for heating homes or generating electricity. - Waste gases must be monitored. - Downside: Carbon dioxide is produced, contributing to climate change.	Collection and transportation costs	- Collecting, transporting, and sorting plastic waste is expensive.
Recycle	- Some plastics, like Polyester PET, are easy to recycle. - Recycling has benefits such as reducing pollution, conserving resources, and saving money.		

* Polyamides (Nylon): Polymers that contain amide linkage (-CONH-) ($\text{C}=\text{O}-\text{N}-\text{H}$)

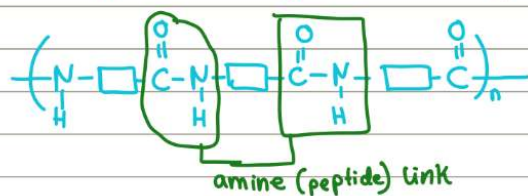
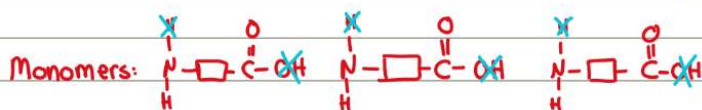
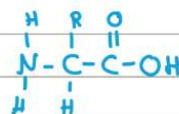
Monomers: -diacid -diamine



Addition polymerisation	Condensation polymerisation
- Many molecules of single monomer - Monomers are unsaturated (alkenes) - Monomers join by opening the $\text{C}=\text{C}$ double bond - Only one product formed: Polymer - Non-biodegradable - Resistant to acids	- Molecules of two monomers - Monomers contain reactive functional group at the end of a molecule - A small molecule is lost (usually water) - Two products formed: - Polymer - Water - Biodegradable - Hydrolysed by acids and enzymes

2- Natural macromolecules: Proteins are polymers build up from amino acids monomers. 20 essential amino

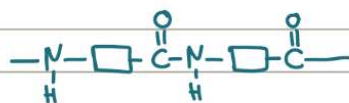
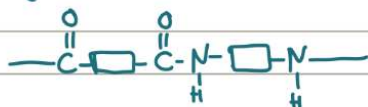
The general structure of an amino acid molecule:



* The difference between nylon and proteins:

Nylon: acid, acid, amine, amine

proteins: amine, acid, amine acid



Nylon	Protein
- Synthetic polyamide - Only two monomers - One monomer is dicarboxylic acid, and the other is diamine → $2\text{C}=\text{O}$, $2\text{N}-\text{H}$	- Natural polyamide - Many different monomers (20 different amino acids) - Each amino acid has an amine group and a carboxylic group on the same carbon → NH_2, COOH

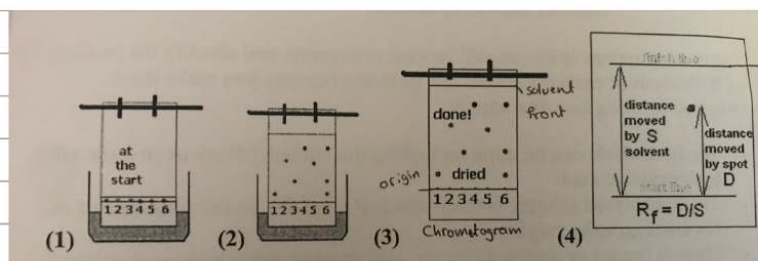
Proteins are broken down by acid hydrolysis: breaking down of molecules with water to form two or more products.

- Hydrolysis is accelerated if the substance is heated with an acid or alkali

- Proteins are broken down into amino acids by enzymes when food is digested.

- Proteins can be hydrolysed into their amino acids by heating them with aqueous hydrochloric acid solution

The mixture of amino acids can then be separated and identified by chromatography.



1- Samples are spotted on start line, the paper is then placed in solvent, but the solvent level must be below the start line of pencil

2- Solvent rises up the paper

3- When the solvent is near the top of the paper, remove paper and dry. For colorless amino acids, spray the paper with a locating agent, which gives a purple spot on each amino acid

4- Measure the R_f values to identify the amino acid

Reference $R_f = \frac{\text{Distance moved by the spot (D)}}{\text{Distance moved by the solvent (S)}}$